

2.9.1

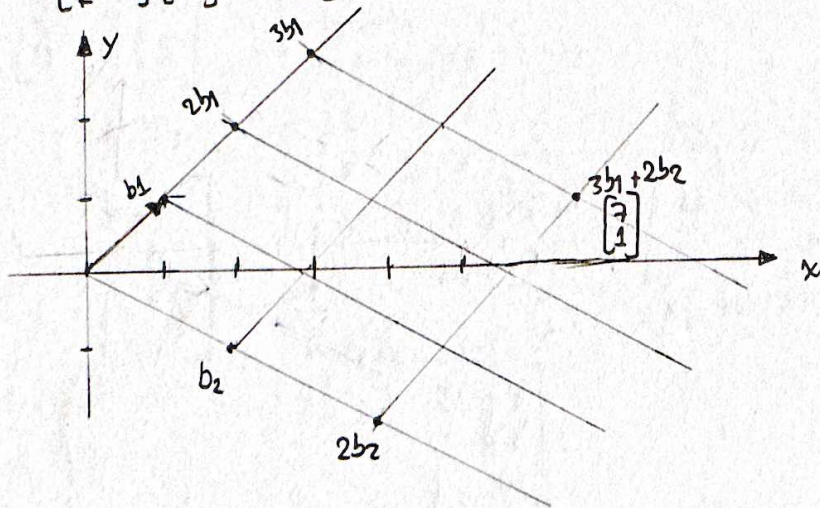
Find the vector $\vec{x}$ determined by the given coordinate vector $[\vec{x}]_\beta$ and the given basis β .

$$\beta = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\}, [\vec{x}]_\beta = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$[\vec{x}]_\beta$ is the coordinate vector of $\vec{x}$ in terms of β .

$$\vec{x} = 3\vec{b}_1 + 2\vec{b}_2 = \begin{bmatrix} 7 \\ 1 \end{bmatrix}, \text{ which also equals}$$

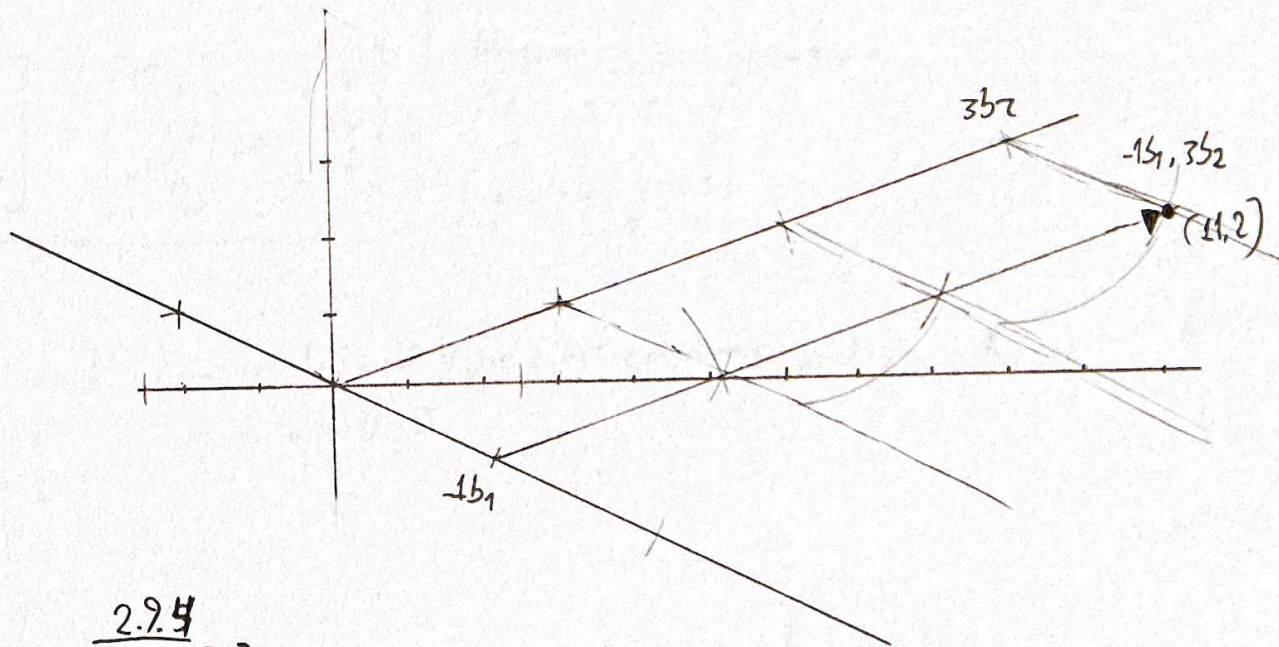
$$\begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$



2.9.2

$$\beta = \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}, [\vec{x}]_\beta = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2+9 \\ -1+3 \end{bmatrix} = \begin{bmatrix} 11 \\ 2 \end{bmatrix}$$



2.9.4

Find $[\vec{x}]_\beta$

$$\vec{b}_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} -4 \\ 7 \end{bmatrix}, \vec{x} = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$$

Find coordinates relative to basis

$$\begin{bmatrix} 1 & -4 & -8 \\ -3 & 7 & 9 \end{bmatrix} \begin{array}{l} r_2 \leftarrow r_2 \cdot (-1/5) \\ r_1 \leftarrow r_1 + 4r_2 \\ \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \end{bmatrix} \end{array}$$

$$r_2 \leftarrow r_2 + 3r_1$$

$$\begin{bmatrix} 1 & -4 & -8 \\ 0 & -5 & -15 \end{bmatrix} \begin{array}{l} [\vec{x}]_\beta = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \end{array}$$

$$4 \begin{bmatrix} 1 \\ -3 \end{bmatrix} + 3 \begin{bmatrix} -4 \\ 7 \end{bmatrix} = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$$

$$[\vec{x}]_\beta = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

2.9.5 Find $[\vec{x}]_\beta$

$$\vec{b}_1 = \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} -3 \\ -7 \\ 5 \end{bmatrix}, \vec{x} = \begin{bmatrix} 4 \\ 10 \\ -7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 4 \\ 5 & -7 & 10 \\ -3 & 5 & -7 \end{bmatrix} \quad [\vec{x}]_\beta = \begin{bmatrix} 1/4 \\ -5/4 \end{bmatrix}$$

$$r_2 \leftarrow r_2 - 5r_1 \\ r_3 \leftarrow r_3 + 3r_1$$

$$\begin{bmatrix} 1 & -3 & 4 \\ 0 & 8 & -10 \\ 0 & -4 & 5 \end{bmatrix}$$

$$r_2 \leftarrow r_2 + 2r_3$$

$$\begin{bmatrix} 1 & -3 & 4 \\ 0 & 0 & 0 \\ 0 & 1 & -5/4 \end{bmatrix}$$

$$r_2 \leftrightarrow r_3$$

$$r_1 \leftarrow r_1 + 3r_2$$

$$\begin{bmatrix} 1 & 0 & 1/4 \\ 0 & 1 & -5/4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$1/4 \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix} - 5/4 \begin{bmatrix} -3 \\ -7 \\ 5 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 5/4 \\ -3/4 \end{bmatrix} + \begin{bmatrix} 15/4 \\ 35/4 \\ -25/4 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ -7 \end{bmatrix}$$

$$r_2 \leftarrow r_2 + 3r_1$$

$$r_3 \leftarrow r_3 - 2r_1$$

$$r_4 \leftarrow r_4 + 4r_1$$

$$\begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 0 & 5 & -7 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 10 & -9 \end{bmatrix}$$

$$r_4 \leftarrow r_4 - 2r_2$$

$$\begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 0 & 5 & -7 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 0 & 1 & -7/5 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$r_1 \leftarrow r_1 - 2r_2$$

$$\begin{bmatrix} 1 & -3 & 0 & -4/5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 4 - x_2 - x_3$$

$$x_2 = 3x_3 + 4 - x_1 = 4 - x_1 = 0$$

$$-x_3 = 0 \Rightarrow x_3 = 0$$

$$x_1 = 4 - x_2 = 4$$

$$x_2 = 0$$

$$x_3 = 0$$

Columns 1, 3 and 4 are pivot columns and form a basis for H . $\dim H = 3$.

2.9.13

Find a basis for the subspace spanned by

$$\begin{bmatrix} 1 \\ -3 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} -3 \\ 9 \\ -6 \\ 12 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -4 \\ 5 \\ -3 \\ 7 \end{bmatrix}$$

Performing row operations

$$\begin{bmatrix} 1 & -3 & 2 & -4 \\ -3 & 9 & -1 & 5 \\ 2 & -6 & 4 & -3 \\ -4 & 12 & 2 & 7 \end{bmatrix}$$

2.9.15

Suppose a 5×8 matrix A has five pivot columns. Is $\text{Col } A = \mathbb{R}^5$? Is $\text{Nul } A = \mathbb{R}^3$?

If A has 8 columns, and five of them are pivot columns, this means 3 of the columns are linear combinations of the others.

And indeed, $\text{Col } A \in \mathbb{R}^5$.

The vectors multiplying the free parameters for a basis for $\text{Nul } A$. Though three-dimensional, $\text{Nul } A$ is not in $\mathbb{R}^3$ because it is a subspace of $\mathbb{R}^8$.

Remember that for an $m \times n$ matrix A , $\text{Nul } A$ is a subspace of $\mathbb{R}^n$.

2.9.31

If possible, construct a 3×4 matrix A such that $\dim \text{Nul } A = 2$ and $\dim \text{Col } A = 2$.

$$\begin{bmatrix} \blacksquare & \blacksquare & \blacksquare & \blacksquare \\ \square & \blacksquare & \blacksquare & \blacksquare \\ \square & \square & \square & \square \end{bmatrix}$$

For $\dim \text{Col } A$ to equal 2, A needs two pivot columns.

To construct such a matrix, create two vectors in $\mathbb{R}^3$ that are obviously linearly independent and place them in a matrix along with a copy of each vector.

For instance

$$\vec{a}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \vec{a}_2 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & -1 & -1 \\ 3 & 3 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 1 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3 \quad x_4$

$x_1, x_3 = \text{pivot}$

$x_2, x_4 = \text{free}$

$$x_1 + x_2 = 0$$

$$x_3 + x_4 = 0$$

$$x_1 = -x_2$$

$$x_3 = -x_4$$

$$\vec{x} = \begin{bmatrix} -x_2 \\ x_2 \\ -x_4 \\ x_4 \end{bmatrix} \quad x_2 = s, x_4 = t \Rightarrow \vec{x} = s \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\vec{u}} + t \underbrace{\begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}}_{\vec{v}}$$

$$\text{Col } A = \text{span}\{\vec{a}_1, \vec{a}_2\}$$

$$\text{Nul } A = \text{span}\{\vec{u}, \vec{v}\}$$